Comparative Analysis of Deep Learning Models for Voxel-Based 3D Spatial Relation Operations

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Abstract

Voxel-based spatial operations are important for three-dimensional spatial information analysis in fields such as digital twins, autonomous driving, and robot navigation. However, conventional mathematical and rule-based approaches have limited flexibility in handling complex 3D structures and diverse spatial relation patterns. In this study, voxel-based 3D spatial operations are formulated as a learning-based spatial relation classification problem. A synthetic voxel dataset was constructed using two 3D objects in a 64 by 64 by 64 voxel grid with six spatial relation classes. Three deep learning models, MLP, 3D CNN, and Voxel Transformer, were applied to evaluate whether spatial relation patterns can be learned from voxel data. The experimental results show that the learning-based models achieved reliable classification performance, suggesting the feasibility of deep learning-based voxel operation techniques.

Keywords— *Voxel, 3D Spatial Relation Operation, Deep Learning, Multi Layer Perceptron, 3D CNN, Voxel Transformer*

SUMMARY OF THIS PAPER

A. Introduction

Recently, the demand for analyzing three-dimensional spatial information has increased in various fields, such as digital twins, autonomous driving, and robot navigation. Accordingly, voxel-based data models, which represent space by dividing it into regular cubic cells, have attracted increasing attention. Voxels can represent various attributes, such as occupancy, semantic information, height, and risk, at the cell level, making them suitable for intuitive and systematic 3D spatial operations.

However, voxel-based spatial operations face several challenges. As the resolution increases, the data size grows rapidly, and spatial contexts such as object boundaries, interior–exterior relationships, adjacency, and connectivity must be considered. In particular, rule-based approaches often have difficulty flexibly handling complex 3D structures and diverse spatial patterns.

In this study, voxel-based 3D spatial operations are formulated as a learning-based spatial relation classification problem. MLP, 3D CNN, and Voxel Transformer models are applied to predict the spatial relation between two voxel objects. The MLP is used as a baseline model based on flattened voxel features, while the 3D CNN learns local spatial patterns among neighboring voxels. The Voxel Transformer captures relationships between sparse voxel tokens through its attention mechanism.

Through this approach, this study examines how effectively artificial intelligence models can learn voxel-based spatial relation patterns and suggests the potential of intelligent voxel operation techniques for future digital twin and 3D spatial information analysis.

B. Materials and Methods

1) Voxel Trainset Construction

First, the input voxel data were generated as $64 \times 64 \times 64$ voxel grids representing two 3D objects, where each voxel value indicates empty space, object A, object B, or the overlapping region. Second, six spatial relation classes were defined: disjoint, meet, contains, within, equal, and overlap. Third, for each generated pair of objects, the actual spatial relation was automatically checked and assigned as a ground truth label using both a class index and one-hot encoding. All samples were generated with a fixed voxel resolution of $64 \times 64 \times 64$, and object sizes were randomly controlled within predefined ranges to produce diverse spatial patterns. Finally, the dataset was divided into 18,000 training samples, 3,000 validation samples, and 3,000 test samples, with each of the six classes generated in a balanced manner.

2) Multi Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) model was used as a baseline model for voxel-based spatial relation classification. Adaptive average pooling was applied to reduce the spatial resolution of the voxel features to $8 \times 8 \times 8$. The pooled voxel features were flattened into a one-dimensional feature vector and passed through fully connected layers. The model consisted of four hidden layers, with 512 units in each layer. The GELU function was used as the nonlinear activation function, and batch normalization and a dropout rate of 0.1 were applied. The final layer produced classification results indicating the spatial relation between the two voxel objects.

3) 3D Convolutional Neural Network (3D CNN)

The 3D Convolutional Neural Network (3D CNN) model was designed to learn local spatial patterns from voxel grids. The model used four 3D convolutional layers with a kernel size of 3×3×3 to extract local features from neighboring voxels. Each convolutional layer was followed by a ReLU activation function and a 3D Max-Pooling layer. The extracted spatial features were used to predict the spatial relationship classes between the objects. Unlike the MLP, the 3D CNN can reflect local spatial relationships such as adjacency, boundary structure, and connectivity. Furthermore, to optimize performance and prevent overfitting, the training pipeline integrated an Adaptive 3D Global Average Pooling (GAP) layer and a cosine annealing learning rate scheduler for stable convergence.

4) Voxel Transformer (VT)

The Voxel Transformer (VT) model was used for voxel-based spatial relation classification. Given a $64 \times 64 \times 64$ categorical voxel grid, the input space was partitioned into $4 \times 4 \times 4$ blocks. Each non-empty block was converted into a sparse token represented by four ratio-based features, namely A_{ratio}, B_{ratio}, $\text{overlap}_{\text{ratio}}$, and $\text{occupancy}_{\text{ratio}}$. As a result, the input was transformed into a $16 \times 16 \times 16$ sparse block grid. The token features were projected into 96-dimensional hidden representations and processed by a VT backbone with four attention heads and stage strides of $[1, 2, 2]$. Each stage consisted of one Sparse Voxel Module and two Submanifold Voxel Modules, incorporating local and dilated attention

with Fast Voxel Query. Finally, the sparse features were aggregated using global mean and max pooling, and the resulting feature vector was passed to a classification head for six spatial relation classes.

5) Research Design

In this study, MLP, 3D CNN, and Voxel Transformer models were applied to classify voxel-based 3D spatial relations, and their performances were compared using the same voxel dataset. The input data consisted of $64 \times 64 \times 64$ voxel grids representing two spatial objects, while the target label indicated one of six spatial relation classes: *disjoint*, *meet*, *contains*, *within*, *equal*, and *overlap*.

The prediction task was defined as a multi-class classification problem that predicts the spatial relation between two voxel objects. Accuracy and F1-score were used as evaluation metrics to compare the overall classification performance of each model. In addition, confusion matrices were analyzed to examine class-wise prediction patterns and identify which spatial relation classes were more difficult to distinguish.

Finally, the performances of the MLP, 3D CNN, and Voxel Transformer models were compared to analyze their learning ability for voxel-based spatial relation classification. Through this comparison, the performance differences among the MLP, which uses flattened voxel features, the 3D CNN, which learns local spatial patterns, and the Voxel Transformer, which models relationships among voxel tokens, were examined.

C. Results and Discussion

1) Performance Comparison

Table 1 presents the performance comparison of the MLP, 3D CNN, and Voxel Transformer (VT) models for voxel-based spatial relation classification.

Table 1. Performance comparison of MLP, 3D CNN, and Voxel Transformer models.

Model	Accuracy	F1-score
MLP	0.8706	0.8696
3D CNN	0.9967	0.9900
VT	**0.9980**	**0.9979**

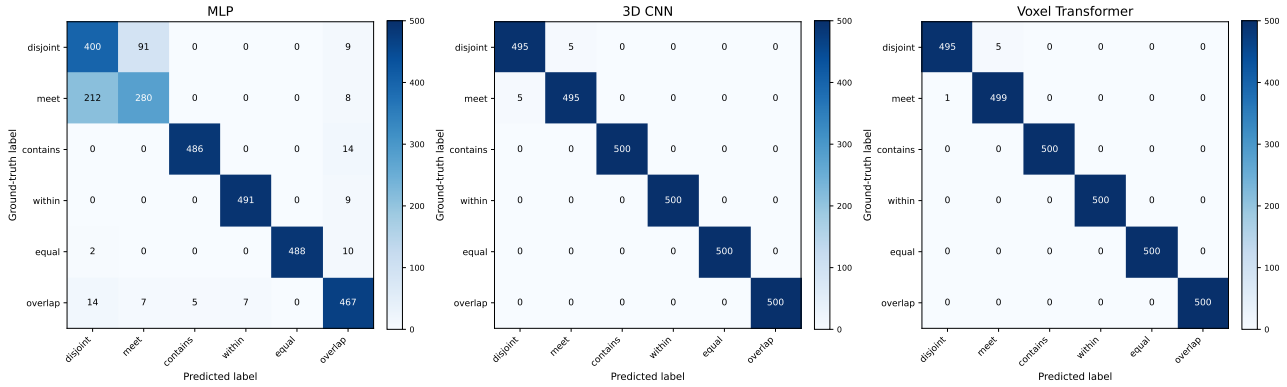


Fig. 1. Confusion matrices of MLP, 3D CNN, and Voxel Transformer models.

2) Discussion on Model Characteristics

The comparison results show that the MLP achieved lower performance than the 3D CNN and Voxel Transformer in terms of accuracy and F1-score. This performance gap is mainly related to the frequent misclassification between the *meet* and *disjoint* classes observed in the confusion matrix. In particular, the MLP often classified *meet* samples as *disjoint*, indicating difficulty in distinguishing boundary contact from complete separation. Both *meet* and *disjoint* have an overlap ratio of zero, which makes them difficult to distinguish when the voxel data is flattened into a one-dimensional feature vector.

In the MLP model, the flattening process does not explicitly preserve the local spatial adjacency structure between voxels. Therefore, the MLP has a limitation in determining whether two objects are completely separated or only touching at their boundaries.

In contrast, the 3D CNN overcomes this limitation by applying convolutional kernels directly to the three-dimensional voxel space. Through this process, the 3D CNN can effectively capture local geometric patterns and neighboring voxel relationships. Similarly, the Voxel Transformer improves classification performance by modeling relationships between voxel tokens while preserving spatial structural information.

As a result, both the 3D CNN and Voxel Transformer showed better performance than the MLP, especially in distinguishing spatially similar classes such as *meet* and *disjoint*. These results indicate that preserving three-dimensional spatial information is important for accurate spatial relation classification.

D. Conclusion

In this study, voxel-based 3D spatial operations were formulated as a learning-based spatial relation classification problem to address the limited flexibility of predefined mathematical and rule-based methods. The experimental results showed that deep learning models can learn spatial relation patterns from voxel data and predict spatial relations with reliable performance.

This indicates that voxel-based spatial operations do not always need to be explicitly defined only by fixed formulas or handcrafted rules. Instead, learning-based models can provide a more flexible approach for handling complex three-dimensional structures and diverse spatial patterns. Therefore, the proposed approach demonstrates the feasibility of intelligent voxel operation techniques for future applications in digital twins, autonomous driving, robot navigation, and 3D spatial information analysis.

As future work, the proposed framework can be extended toward distance-aware and topology-aware voxel reasoning. Incorporating object distance, boundary contact, and voxel adjacency may improve the classification of spatially similar relations such as *meet* and *disjoint*. Graph-based models can also be explored to represent voxel groups and their spatial relationships more explicitly.